

PRACTICAL NO: 13

AIM: To simulate (a) a single mote, (b) mote-to-mote radio communication, analysing signal strength and packet loss under varying radio gain values.

nano single.tcl

```
it-03@DESKTOP-FL4Q95A: ~/practical
GNU nano 8.7.1 single.tcl
Phy/WirelessPhy set Pt_ 0.0001

set ns [new Simulator]
set tf [open single.tr w]
$ns trace-all $tf

set topo [new Topography]
$topo load_flatgrid 100 100
create-god 1
$ns node-config -adhocRouting DSDV \
    -llType LL -macType Mac/802.11 \
    -ifqType Queue/DropTail/PriQueue -ifqLen 50 \
    -antType Antenna/OmniAntenna \
    -propType Propagation/TwoRayGround \
    -phyType Phy/WirelessPhy -channelType Channel/WirelessChannel \
    -topoInstance $topo -agentTrace ON -routerTrace ON -macTrace ON

set m0 [$ns node]
$m0 set X 10
$m0 set Y 10
$m0 set Z 0

set udp [new Agent/UDP]
$ns attach-agent $m0 $udp
set null [new Agent/Null]
$ns attach-agent $m0 $null
$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set packetSize 36
$cbr set interval 1
$ns at 1.0 "$cbr start"
$ns at 20.0 "$cbr stop"
$ns at 21.0 "$ns halt"
$ns run
```

```
it-03@DESKTOP-FL4Q95A:~/practical$ head -3 single.tcl
Phy/WirelessPhy set Pt_ 0.0001

set ns [new Simulator]
it-03@DESKTOP-FL4Q95A:~/practical$ ns single.tcl
num_nodes is set 1
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 75.5
SORTING LISTS ...DONE!
it-03@DESKTOP-FL4Q95A:~/practical$ (ns single.tcl && head
```

```
it-03@DESKTOP-FL4Q95A:~/practical$ grep "^s" single.tr | grep AGT | head
s 1.000000000 _0_ AGT --- 1 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [0] 0 0
s 2.000000000 _0_ AGT --- 2 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [1] 0 0
s 3.000000000 _0_ AGT --- 3 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [2] 0 0
s 4.000000000 _0_ AGT --- 4 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [3] 0 0
s 5.000000000 _0_ AGT --- 5 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [4] 0 0
s 6.000000000 _0_ AGT --- 6 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [5] 0 0
s 7.000000000 _0_ AGT --- 7 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [6] 0 0
s 8.000000000 _0_ AGT --- 8 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [7] 0 0
s 9.000000000 _0_ AGT --- 9 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [8] 0 0
s 10.000000000 _0_ AGT --- 10 cbr 36 [0 0 0 0] ----- [0:0 0:1 32 0] [9] 0 0
it-03@DESKTOP-FL4Q95A:~/practical$ grep "^s" single.tr | grep -c AGT
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it-03@DESKTOP-FL4Q95A:~/practical$ nano single.tcl
```

(b) mote-to-mote radio communication, analysing signal strength and packet loss under varying radio gain values

nano serial.tcl

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```
it-03@DESKTOP-FL4Q95A: ~/practical
GNU nano 8.7.1 serial.tcl
set ns [new Simulator]
set tf [open serial.tr w]
$ns trace-all $tf
set nf [open serial.nam w]
$ns namtrace-all $nf

set mote [$ns node]
set pc [$ns node]
$ns duplex-link $mote $pc 115200b 1ms DropTail

set udp [new Agent/UDP]
$ns attach-agent $mote $udp
set sink [new Agent/Null]
$ns attach-agent $pc $sink
$ns connect $udp $sink

set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set packetSize_ 36
$cbr set interval_ 0.5
$ns at 0.5 "$cbr start"
$ns at 20 "$cbr stop"
$ns at 21 "$ns halt"
$ns run_

it-03@DESKTOP-FL4Q95A:~/practical$ nano serial.tcl
it-03@DESKTOP-FL4Q95A:~/practical$ head -3 serial.tcl
set ns [new Simulator]
set tf [open serial.tr w]
$ns trace-all $tf
it-03@DESKTOP-FL4Q95A:~/practical$ ns serial.tcl
it-03@DESKTOP-FL4Q95A:~/practical$ grep "^r" serial.tr | head -20
r 0.5035 0 1 cbr 36 ----- 0 0.0 1.0 0 0
r 1.0035 0 1 cbr 36 ----- 0 0.0 1.0 1 1
r 1.5035 0 1 cbr 36 ----- 0 0.0 1.0 2 2
r 2.0035 0 1 cbr 36 ----- 0 0.0 1.0 3 3
r 2.5035 0 1 cbr 36 ----- 0 0.0 1.0 4 4
r 3.0035 0 1 cbr 36 ----- 0 0.0 1.0 5 5
r 3.5035 0 1 cbr 36 ----- 0 0.0 1.0 6 6
r 4.0035 0 1 cbr 36 ----- 0 0.0 1.0 7 7
it-03@DESKTOP-FL4Q95A:~/practical$ grep -c "^r" serial.tr
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```

Mote.ned:

```
package motecommunication;
```

```
simple Mote
```

```
{
    parameters:
        double sendInterval @unit(s);
    gates:
        input in;
        output out;
}
```

RadioNetwork.ned:

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```
package motecommunication;

network RadioNetwork
{
    submodules:
        mote1: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        mote2: Mote
        {
            parameters:
                sendInterval = 0s;
        }

    connections:
        mote1.out --> mote2.in;
        mote2.out --> mote1.in;
}
```

OUTPUT:

